B.Sc. BOTANY — YEAR I

MAJOR COURSE III

APPLIED BOTANY

UNIT V — Plant Tissue Culture, Recombinant DNA and Bioinformatics

Detailed Teacher-Style Notes

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

UNIT V — BIOTECHNOLOGY AND BIOINFORMATICS

Syllabus topics covered: plant tissue culture—definition, types and importance; recombinant technique—introduction, tools, technique and importance; bioinformatics—definition, concept, tools; introduction to BLAST and FASTA; importance of bioinformatics; Indian scientists' contribution in plant tissue culture; applications of BLAST and FASTA.

5.1 Plant tissue culture

Definition — Plant tissue culture: The aseptic in-vitro culture of plant cells, tissues or organs on a defined nutrient medium under controlled environmental conditions.

Scientific basis: plant cells can show **totipotency**, meaning that under appropriate conditions some living plant cells can regenerate a complete plant. Actual regeneration depends on genotype, explant, medium and culture conditions.

Standard plant tissue-culture workflow

Select explant → Surface sterilize → Aseptic in oculation → Culture on medium → Callus / o rganogenes is / → Rooting and hardening → Field transfer

Types / major culture forms

Type	Core idea	Use
Callus culture	Unorganized cell mass grown in vitro	Regeneration and experimental studies
Organ culture	Culture of organs or organ parts	Developmental and propagation studies
Meristem culture	Culture of shoot-tip/meristem tissue	Rapid multiplication and clean stock programmes
Embryo culture	Culture of isolated embryo	Rescue and developmental studies
Cell suspension culture	Cells grown in liquid medium	Mass culture and metabolite studies
Anther/microspore culture	Culture of male reproductive tissues	Haploid and doubled-haploid work

Importance

- Rapid clonal multiplication of selected plants.
- Production of uniform planting material under controlled conditions.
- Conservation and propagation of valuable germplasm.
- Support for breeding, transformation and molecular studies.
- Production of specialized metabolites in selected culture systems.

5.2 Recombinant DNA technique

Definition — Recombinant DNA technology: A set of molecular methods used to construct and work with DNA molecules formed by joining genetic material from different sources or by designed manipulation of DNA sequences.

Core tools

Tool	Role
Restriction endonucleases	Cut DNA at specific recognition sequences
DNA ligase	Joins compatible DNA fragments
Vectors	Carry inserted DNA into a host system
Host cells	Replicate or express introduced genetic material
Polymerases and related enzymes	Support DNA synthesis and amplification in relevant workflows
Selectable markers / screening tools	Help identify cells carrying the desired construct

Simplified recombinant-DNA workflow

Importance

- Gene-function studies.
- Production of recombinant biomolecules.
- Plant genetic improvement research.
- Molecular diagnostics and biotechnology.
- Foundation for many modern genetic engineering applications.

5.3 Bioinformatics

Definition — Bioinformatics: An interdisciplinary field that uses computational methods, databases and algorithms to store, retrieve, analyze and interpret biological information such as DNA, RNA and protein sequences.

Bioinformatics pipeline

Common tool categories

Category	Example function
Sequence databases	Store and retrieve nucleotide/protein sequences
Sequence similarity tools	Compare a query sequence with database sequences
Alignment tools	Identify similarities and differences between sequences
Genome browsers	Explore genomic regions and annotations
Phylogenetic tools	Infer relationships from sequence data
Visualization tools	Display sequences, structures or relationships

5.4 BLAST and FASTA

Definition — BLAST: Basic Local Alignment Search Tool; a widely used family of programs for finding local sequence similarity between a query sequence and sequences in a database.

Definition — FASTA: A sequence-comparison method and, in common usage, a simple text-based sequence format in which a header line begins with the character > followed by sequence data.

Feature	BLAST	FASTA / FASTA-style search concept
Main purpose	Rapid local similarity search	Sequence representation and similarity-search framework
Typical input	Nucleotide or protein query	Nucleotide or protein sequence
Main output	Database matches with alignment statistics	Similarity matches/alignments depending on implementation
Use	Identify homologous or similar sequences	Store/compare sequence information

Applications of BLAST and FASTA

• Identify similar sequences and possible homologs.

• Check whether a sequence resembles known genes or proteins.

• Support annotation of newly obtained sequences.

• Compare plant, microbial or other biological sequences.

• Generate hypotheses for further experimental validation.

5.5 Indian scientists' contribution in plant tissue culture

The syllabus explicitly asks for this activity. For a final submitted answer, use contributions supported by your prescribed textbook/teacher or a verified institutional source. The syllabus page itself does not name specific scientists, so unsupported names or achievements should not be invented in the notes.

Exam focus: Unit V high-yield: definition and workflow of plant tissue culture; totipotency; culture types; recombinant-DNA tools and steps; bioinformatics definition; BLAST vs FASTA comparison and applications.